JSPM's RAJARSHI SHAHU COLLEGE OF ENGINEERING
(An Autonomous Institute, Affiliated to Savitribai Phule Pune university)
Department of Electronics & Telecommunication Engineering
One Page Project Summary (A.Y 2026-27)

PROJECT TITLE: Multimodal AI Teaching Copilot for Real-Time Visual Learning Generation

CATEGORY OF THE PROJECT: Design and Development of Software

PROJECT DOMAIN: Artificial Intelligence in Education (speech recognition, large language models, real-time visual systems)

GROUP NO: _____

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ABSTRACT:

Prepared slides cannot show the examples and explanations a teacher adds while teaching, and an LLM that regenerates slides from a growing transcript keeps rewriting what students are reading. This project builds a teaching copilot that listens to the teacher and, during the lecture, puts a structured, continuity-aware slide deck on the classroom projector: typed layouts, typeset formulas, checked images and live concept simulations. Speech is transcribed locally; a large language model only interprets bounded units of speech, while a deterministic planner decides the slides and only extends the deck. The system runs on one laptop with a 4 GB GPU and free AI services.

PROBLEM STATEMENT:

To design a low-cost system that converts a teacher's live speech into a structured, stable and transcript-faithful visual lesson (slides, formulas, images and parametric simulations) on the classroom projector during the lecture, while keeping the teacher in control.

OBJECTIVES:

- Capture and transcribe classroom speech locally in real time (Silero VAD, faster-whisper on GPU).
- Track the topic online and interpret bounded speech units with an LLM that returns only validated labels.
- Plan slides deterministically so the deck only grows; guard content against the transcript.
- Show slides, formulas, images and spoken-value-driven simulations on projector, teacher and student views.
- Evaluate latency, accuracy, stability and teacher ratings on held-out lectures and against an LLM baseline.

IMPLEMENTATION:

Python (asyncio) core with typed events and an SQLite event log; faster-whisper (Whisper medium.en, int8) and Silero VAD for speech; MiniLM embeddings for topic tracking; a rule-based gate and bounded prompts (< 3,000 tokens) to the Groq free tier (gpt-oss-120b, Qwen3) with Pydantic validation and a deterministic fallback; extend-only planner, coverage and formula guards; composer with 10+ layouts, KaTeX and CLIP-checked open-licence images; FastAPI + WebSocket views /display, /control and /view; 21 simulation templates (mechanics, circuits, RLC, p-n junction, TCP, routing, optics, gas laws, biological processes) with Python solvers; post-lecture notes, assignments, quizzes and PPTX.

RESULTS:

- 36 held-out lectures (409 min, 6 speakers + MIT OCW / NPTEL): median speech-to-slide delay **11.2 s**.
- 18 annotated lectures: topic-boundary precision **0.78**, **86.6 %** of key facts on a slide, only **0.9 %** statements not said; five teachers rated the slides **3.87–4.17 / 5**.
- Versus an LLM-controlled baseline: seen statements rewritten **2.0 % vs 38.2 %**, lower in all 18 lectures ($p < 0.001$).
- Simulations: show precision and recall **1.00** on 66 lectures, no false simulation; solver values match hand-computed textbook values; 0 LLM tokens.
- Tests: 1,547 unit tests and 122 real-browser tests passing.

SPONSORSHIP:

Nil (self-funded; only free and open-source tools used).

PROJECT COMPETITION: —

PUBLICATION:

IEEE conference paper “Multimodal AI Teaching Copilot: Live Slide Generation from Lecture Speech” (P. Potdar, P. Raut, D. Magdum, S. Kale, S. Patnaik), 6 pages. Conference / status: _____

SPECIAL ACHIEVEMENT IF ANY: —

Mrs. Deepali Magdum	Mrs. D. K. Magdum, Mrs. S. P. Pattanaik	Dr. S. C. Wagaj
Project Guide	**Project Coordinators**	**HOD**